```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
```

```
import pandas as pd
```

```
pd.__version__
```

```
'2.2.2'
```

```
df = pd.DataFrame([11,22,33], columns=['Col_Name'])
print(df)
```

```
   Col_Name
0        11
1        22
2        33
```

```
print(type(df))
```

```
<class 'pandas.core.frame.DataFrame'>
```

using dictionary of lists

```
data = {
    'Name' : ['Madhav','vishika','Lalita','Rishabh'],
    'Age' : [16,17,18,19],
    'Salary' : [9000,7000,5000,3000]
}
```

```
df = pd.DataFrame(data)
print(df)
```

```
      Name  Age  Salary
0   Madhav   16    9000
1  vishika   17    7000
2   Lalita   18    5000
3  Rishabh   19    3000
```

```
print(type(df))
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
df.head(2)
```

	Name	Age	Salary
0	Madhav	16	9000
1	vishika	17	7000

Next steps: Generate code with df New interactive sheet

```
df.tail(2)
```

	Name	Age	Salary
2	Lalita	18	5000
3	Rishabh	19	3000

```
df[['Name', 'Monthly_Salary']]
```

	Name	Monthly_Salary
0	Madhav	9000
1	vishika	7000
2	Lalita	5000
3	Rishabh	3000

```
df[['Name']]
```

	Name
0	Madhav
1	vishika
2	Lalita
3	Rishabh

```
df.shape
```

```
(4, 3)
```

```
df.columns
```

```
Index(['Name', 'Age', 'Salary'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Name            4 non-null      object
 1   Age             4 non-null      int64
 2   Monthly_Salary  4 non-null      int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
```

```
df.describe()
```

	Age	Monthly_Salary
count	4.000000	4.000000
mean	17.500000	6000.000000
std	1.290994	2581.988897
min	16.000000	3000.000000
25%	16.750000	4500.000000
50%	17.500000	6000.000000
75%	18.250000	7500.000000
max	19.000000	9000.000000

```
df.rename(columns={'Salary': 'Monthly_Salary'},inplace=True)
```

```
df
```

	Name	Age	Monthly_Salary
0	Madhav	16	9000
1	vishika	17	7000
2	Lalita	18	5000
3	Rishabh	19	3000

Next steps: Generate code with df New interactive sheet

```
df.to_csv('test_data.csv',index=False)
load.df=pd.read_csv('test_data.csv')
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_194/3721901547.py in <cell line: 0>()
      1 df.to_csv('test_data.csv',index=False)
----> 2 load.df=pd.read_csv('test_data.csv')

NameError: name 'load' is not defined
```

Next steps: Explain error

rows and columns selection

```
df[['Name','Monthly_Salary']]
df.loc[df.Name=='Mahadev']
df.loc[df.Name=='Mahadev']&df.loc[df.Monthly_Salary>=50000]
df.loc[0:2]
df.iloc[0:2]
```

	Name	Age	Monthly_Salary
0	Madhav	16	9000
1	vishika	17	7000

filter dataframe filtering columns

```
df
```

	Name	Age	Monthly_Salary
0	Madhav	16	9000
1	vishika	17	7000
2	Lalita	18	5000
3	Rishabh	19	3000

Next steps: Generate code with df New interactive sheet

```
df_age_filter = df[df['Age'] >= 18]
```

```
df_age_filter
```

	Name	Age	Monthly_Salary
2	Lalita	18	5000
3	Rishabh	19	3000

Next steps: Generate code with df_age_filter New interactive sheet

```
df[(df['Age'] >= 18) & (df['Monthly_Salary'] >= 5000)]
```

	Name	Age	Monthly_Salary
2	Lalita	18	5000

```
df_age_filter
```

	Name	Age	Monthly_Salary
2	Lalita	18	5000
3	Rishabh	19	3000

Next steps: Generate code with df_age_filter New interactive sheet

```
df.where(df['Age'] >= 18)
```

	Name	Age	Monthly_Salary
0	NaN	NaN	NaN
1	NaN	NaN	NaN
2	Lalita	18.0	5000.0
3	Rishabh	19.0	3000.0

```
df.where(df['Age'] >= 18, other = 'Not Eligible')
```

	Name	Age	Monthly_Salary
0	Not Eligible	Not Eligible	Not Eligible
1	Not Eligible	Not Eligible	Not Eligible
2	Lalita	18	5000
3	Rishabh	19	3000

Rows and columns - operation(Add,update,delete)

```
df
```

	Name	Age	Monthly_Salary
0	Madhav	16	9000
1	vishika	17	7000
2	Lalita	18	5000
3	Rishabh	19	3000

Next steps: Generate code with df New interactive sheet

```
df['Team'] = ['CEO','HR','CTO','DA']
df
```

	Name	Age	Monthly_Salary	Team
0	Madhav	16	9000	CEO
1	vishika	17	7000	HR
2	Lalita	18	5000	CTO
3	Rishabh	19	3000	DA

Next steps: Generate code with df New interactive sheet

```
df['Bonus'] = df['Monthly_Salary'] * 0.20
df
```

	Name	Age	Monthly_Salary	Team	Bonus
0	Madhav	16	9000	CEO	1800.0
1	vishika	17	7000	HR	1400.0
2	Lalita	18	5000	CTO	1000.0
3	Rishabh	19	3000	DA	600.0

Next steps: Generate code with df New interactive sheet

```
df.loc[len(df)] = ['ABC', 21, 2100, 'IT', 2000]
df
```

	Name	Age	Monthly_Salary	Team	Bonus
0	Madhav	16	9000	CEO	1800.0
1	vishika	17	7000	HR	1400.0
2	Lalita	18	5000	CTO	1000.0
3	Rishabh	19	3000	DA	600.0
4	ABC	21	2100	IT	2000.0

Next steps: Generate code with df New interactive sheet

```
df.loc[0,'Monthly_Salary'] = 95000
df
```

	Name	Age	Monthly_Salary	Team	Bonus
0	Madhav	16	95000	CEO	1800.0
1	vishika	17	7000	HR	1400.0
2	Lalita	18	5000	CTO	1000.0
3	Rishabh	19	3000	DA	600.0
4	ABC	21	2100	IT	2000.0

Next steps: Generate code with df New interactive sheet

```
df.loc[df.Name =='Madhav','Monthly_Salary'] = 9000
df
```

	Name	Age	Monthly_Salary	Team	Bonus
0	Madhav	16	9000	CEO	1800.0
1	vishika	17	7000	HR	1400.0
2	Lalita	18	5000	CTO	1000.0
3	Rishabh	19	3000	DA	600.0
4	ABC	21	2100	IT	2000.0

Next steps: Generate code with df New interactive sheet

```
df = df.drop(df[df.Name == 'ABC'].index)
```

```
df.drop(1, axis=0)
```

	Name	Age	Monthly_Salary	Team	Bonus
0	Madhav	16	9000	CEO	1800.0
2	Lalita	18	5000	CTO	1000.0
3	Rishabh	19	3000	DA	600.0

```
df.drop('Bonus', axis=1, inplace=True)
df
```

	Name	Age	Monthly_Salary	Team
0	Madhav	16	9000	CEO
1	vishika	17	7000	HR
2	Lalita	18	5000	CTO
3	Rishabh	19	3000	DA

Next steps: Generate code with df New interactive sheet

```
df.rename(columns={'Monthly_Salary': 'Salary'}, inplace=True)
```

```
df.sort_values('Salary')
```

	Name	Age	Salary	Team
3	Rishabh	19	3000	DA
2	Lalita	18	5000	CTO
1	vishika	17	7000	HR
0	Madhav	16	9000	CEO

```
df.sort_values('Salary', ascending=False)
```

	Name	Age	Salary	Team
0	Madhav	16	9000	CEO
1	vishika	17	7000	HR
2	Lalita	18	5000	CTO
3	Rishabh	19	3000	DA

working with date value

```
df['DOJ'] = ['2024-01-01', '2024-01-15', '2024-03-28', '2024-03-03']
df
```

	Name	Age	Salary	Team	DOJ
0	Madhav	16	9000	CEO	2024-01-01
1	vishika	17	7000	HR	2024-01-15
2	Lalita	18	5000	CTO	2024-03-28
3	Rishabh	19	3000	DA	2024-03-03

Next steps: Generate code with df New interactive sheet

```
df['DOJ'].dtype
```

```
dtype('O')
```

```
df['DOJ'] = pd.to_datetime(df['DOJ'])
df['DOJ'].dtype
```

```
dtype('<M8[ns]')
```

Start coding or generate with AI.